

GREEN INDUSTRIES

Green industries courses offer students opportunities to investigate how to care for and sustainably manage our natural and living resources. Students will explore a variety of areas within green industries, including agriculture, floristry, forestry, horticulture, and/or landscaping.

Students will gain practical skills and knowledge by completing a variety of authentic, industry-relevant activities and projects, such as plant propagation and greenhouse maintenance activities; landscape design and planning exercises; assignments that include farm or forestry management plans; and various construction projects. In addition, students will be introduced to concepts pertaining to biodiversity, environmental sustainability, and natural versus artificial products. Students will also investigate the numerous and varied ways in which green industries and society are interdependent.

Green industries courses will prepare students for working safely in the many fields that are encompassed in this subject area. Students will learn about career opportunities in the green industries and about the Essential Skills and work habits that are important for success in these fields.

The list of approved emphasis areas for green industries can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Green Industries, Grade 12

Workplace Preparation

THJ4E

This course enables students to gain further experience with a variety of industry procedures and operations and to acquire additional industry-specific skills. Students will study more complex processes, develop more advanced design and maintenance skills, and explore ways of enhancing environmental sustainability. They will also examine social and economic issues related to the green industries, learn about safe and healthy working practices, study industry standards and codes, and explore career opportunities in the various industries. The knowledge and skills acquired in this course will prepare students for the workplace and apprenticeship training.

Prerequisite: Green Industries, Grade 11, Workplace Preparation

A. GREEN INDUSTRY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of species classification and identification and of relationships between species and geographical regions;
- A2.** demonstrate an understanding of the effects of biotic and abiotic factors on growth and product quality;
- A3.** demonstrate an understanding of designs and processes related to a variety of green industry applications;
- A4.** use mathematical, documentation, research, and communication skills as they apply to the green industries.

SPECIFIC EXPECTATIONS

A1. Species Classification and Geographical Regions

By the end of this course, students will:

- A1.1** use common classification schemes and key identification characteristics to distinguish between different plant and/or animal groups (*e.g., annuals versus perennials, deciduous versus coniferous trees and shrubs, oaks versus birches, Holsteins versus Ayrshires*);
- A1.2** identify geographical regions on the basis of classification criteria relevant to the green industries (*e.g., plant hardiness, growing degree days, elevation, soil type, soil moisture*), and describe how geographical factors determine the distribution of species in these regions;
- A1.3** compare different kinds of ecosystems in terms of their biodiversity (*e.g., a climax forest versus a rejuvenated forest, natural versus managed land, a cultivated field versus a greenhouse*).

A2. Factors Affecting Growth and Product Quality

By the end of this course, students will:

- A2.1** describe the effects of abiotic factors (*e.g., light, temperature, soils, nutrients, topography, moisture, climate change, ethylene gas*) on plant and/or animal growth and post-harvest quality (*e.g., a southern exposure may increase yields by increasing available light; too little moisture may stunt growth*

and reduce yields; too much moisture may encourage growth of mould and mildew; ethylene gas acts as a ripening agent for picked fruits);

- A2.2** describe the effects of biotic factors (*e.g., pests, diseases, weeds*) on plant and/or animal growth and post-harvest quality (*e.g., form defects, stunted growth, reduced yields, damaged fruit*);
- A2.3** describe the effects of interactions between abiotic, biotic, and cultural factors in a variety of environments (*e.g., gardens, greenhouses, barns, florists' coolers, fields, forest stands*);
- A2.4** describe a variety of integrated pest management techniques (*e.g., cultural [tilling and mulching], physical [crop rotation], environmental [introduction of beneficial insects], biological [fungi, nutrients], chemical [pheromones, chemical pesticides]*), and identify situations in which they can be applied effectively.

A3. Designs and Processes

By the end of this course, students will:

- A3.1** describe how design or planning processes are used in a variety of green industry applications (*e.g., preparation of species prescriptions, crop rotation plans, environmental assessments, site layouts, event plans*);

A3.2 explain processes and techniques relating to the propagation, maintenance, and care of plants and/or animals (*e.g., animal breeding, taking cuttings, seeding, irrigation, pruning, clipping, feeding, clearing*);

A3.3 describe new technologies that are being applied to the green industries (*e.g., computerized inventory systems, automated watering systems, aerial and satellite imaging*).

A4. Technological and Mathematical Literacy and Communication Skills

By the end of this course, students will:

A4.1 demonstrate an understanding of terminology used in the green industries and use it correctly in oral and written communication (*e.g., board feet, petiole, focalization, terrace, humus*);

A4.2 use effective documentation practices to record and track important information (*e.g., preparing estimates, recording costs, recording crop yields, preparing inventories, completing equipment maintenance records*);

A4.3 identify technical issues of current significance in the green industries, using information from general media sources (*e.g., newspapers, magazines, the Internet*);

A4.4 perform measurements and calculations accurately for a variety of green industry applications (*e.g., determining basal area, animal weight, crown size, slope; invoicing; surveying*).

B. GREEN INDUSTRY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** demonstrate an understanding of and apply design and production practices that are commonly used in the green industries;
- B2.** apply management strategies for assessing and controlling biotic and abiotic factors that affect plant and/or animal quality;
- B3.** demonstrate competence in technical skills relating to specific applications and tasks in the green industries.

SPECIFIC EXPECTATIONS

B1. Design and Production

By the end of this course, students will:

- B1.1** develop a design or process for a green industry application (*e.g., a landscape design, a crop production plan, a herd management procedure, a plant propagation schedule*);
- B1.2** devise an effective management plan or site layout for a specific application (*e.g., a forest prescription, a herd management plan, an event plan, a landscape construction plan, a block layout and harvesting plan, animal housing*);
- B1.3** demonstrate an understanding of and apply techniques related to the propagation, maintenance, and post-harvest handling of a variety of species (*e.g., pruning, scarification of forest floor, artificial insemination, transplanting large trees, cut flower conditioning*);
- B1.4** demonstrate an understanding of and apply designs and production processes that promote biodiversity, increase ecosystem function, and reduce maintenance requirements (*e.g., mulching, sustainable water gardening, rooftop gardening, naturalizing landscapes*);
- B1.5** demonstrate an understanding of and apply marketing techniques that are commonly used in the green industries (*e.g., product displays, flyer advertising, promotional websites*).

B2. Plant and Animal Management Strategies

By the end of this course, students will:

- B2.1** perform a variety of procedures (*e.g., pest counts, microscopic investigations, visual inspections, estrous cycle monitoring*) to assess biotic conditions that affect plant and/or animal quality;
- B2.2** perform a variety of procedures (*e.g., soil and air temperature measurement, water analysis, form defects analysis, air quality assessment, nutritional assessment, monitoring ethylene gas concentrations*) to assess or measure abiotic conditions that affect plant and/or animal quality;
- B2.3** apply techniques for controlling pests and disorders of plants and/or animals (*e.g., fogging, density planting, encouraging beneficial insects, constructing barriers, setting live traps, inoculations, animal tagging*);
- B2.4** demonstrate an understanding of and apply management techniques that enhance environmental sustainability within the green industries (*e.g., sustainable herd management practices, measures that enhance forest succession, preferential use of native species*).

B3. Technical Skills

By the end of this course, students will:

- B3.1** demonstrate competence in the technical skills required to complete a variety of tasks or

processes in the green industries (*e.g., tractor operation, use of hydraulics, mason saw operation, tree removal, pruning, floral mechanics, operation of computerized feeding equipment*);

B3.2 construct a variety of structures or products that are used in the green industries (*e.g., arbours, boardwalks, retaining walls, raised beds, skid trails, biogas generator*);

B3.3 demonstrate competence in related technical skills required for success within a variety of green industries (*e.g., construction, manufacturing, plumbing, wiring and electrical repair, diesel engine operation, graphic communication, artistic design*);

B3.4 demonstrate competence in specialized technical skills required for unique applications (*e.g., working at heights, water gardening, breeding animals, grafting, creating new convention designs, using specialized software applications*).

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** assess the impact of the green industries on the environment and describe ways of enhancing environmental sustainability;
- C2.** describe social, economic, and cultural relationships involving the green industries.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** assess the environmental sustainability of various practices and procedures used in the green industries (*e.g., harvesting methods, wood product manufacturing methods, naturalized landscaping, global floral sourcing, environmental farm plans, crop rotation, large-scale farming*);
- C1.2** describe ways of reducing negative or enhancing positive environmental consequences through the use of particular materials, products, processes, and disposal methods (*e.g., nutrient recycling, spot spraying for insects and fungus, using organic fertilizer, composting, xeriscaping*);
- C1.3** describe methods used in the green industries to balance economic sustainability with environmental responsibilities (*e.g., selective breeding, selective cutting, organic production methods, environmental best management practices*);
- C1.4** describe the benefits of alternative practices that reduce the environmental impact of green industry operations (*e.g., living walls, natural-scaping, forest certification, tree marking guidelines, fibre crops, armatures and grid work, alternative animal housing systems*);
- C1.5** identify legislation, regulations, standards, and guidelines relating to environmental protection that affect operations in the green industries (*e.g., Clean Water Act, Nutrient Management Act, species importation regulations, tree-cutting by-laws, pest control regulations*).

C2. Technology and Society

By the end of this course, students will:

- C2.1** describe ways in which the green industries are affected by the social, economic, and cultural characteristics of the communities in which they operate (*e.g., relations with Aboriginal communities, ethnic preferences and demands for specialized food products, income distribution and demand for landscaping services and horticultural products, age profile of community and requirements for different types of outdoor recreation facilities*);
- C2.2** identify related industries and describe how they and the green industries are interdependent (*e.g., small-engine manufacturing and lawn care, nutrient research and animal/crop production, equipment manufacturing and farming*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and apply safe working practices as they relate to the green industries;
- D2.** demonstrate an understanding of the business and regulatory environment of the green industries;
- D3.** identify careers in the green industries, and describe the skills, education, and training required for entry into these occupations.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** identify common industry hazards (e.g., *ergonomic hazards, mechanical hazards, chemical hazards*) and sources of information about accident prevention (e.g., *Workplace Hazardous Materials Information System [WHMIS], Passport to Safety*);
- D1.2** identify the personal protective clothing and equipment needed to perform various green industry tasks safely, and use as required to ensure their own and others' safety in the work environment (e.g., *safety glasses, hard hats, safety boots, fall protection gear*);
- D1.3** demonstrate the ability to make appropriate safety decisions that take account of environmental conditions and worker preparedness (e.g., *level of training or experience, access to seasonal work wear, comfort level around large animals, fire response capabilities, weather and site hazards*);
- D1.4** demonstrate an understanding of and apply safe procedures for using and maintaining materials, tools, and equipment (e.g., *avoid moving parts and pinch points; perform a circle check of vehicles and equipment; check condition of materials, hydraulic systems, and protective equipment; check oil and fuel levels*);
- D1.5** demonstrate an understanding of and apply safe procedures for handling plants and/or animals (e.g., *dethorning plants, using ergonomic lifting techniques or devices, using hand protection, securing loads correctly for transport, understanding animal perception, avoiding actions that startle animals, using chutes and restraining devices*);

- D1.6** demonstrate an understanding of specific components of the Occupational Health and Safety Act and other safety-related requirements that relate to the green industries (e.g., *Ontario Forestry Safe Workplace Association guidelines, Grower Pesticide Safety Course certificate, Forest Fires Prevention Act*).

D2. Business and Regulatory Environment

By the end of this course, students will:

- D2.1** identify a variety of industry-related organizations (e.g., *local growers' associations, government departments, non-governmental organizations*), and demonstrate an understanding of their role with respect to the green industries;
- D2.2** identify basic economic concepts and principles that affect operations in the green industries (e.g., *cost of production and transportation, supply and demand*);
- D2.3** outline and comply with legislation and guidelines governing the quality and safety of green industry products and services (e.g., *product quality regulations, grading standards, inspection requirements, voluntary guidelines such as the Landscape Ontario guidelines for the landscaping industry*).

D3. Career Opportunities

By the end of this course, students will:

- D3.1** identify careers in the green industries (e.g., *arborist, florist, herder, greenhouse worker, forester*), and describe the nature and scope of the work involved;

- D3.2** describe the education, training, and certification required for employment in various occupations in the green industries;
- D3.3** describe areas of specialization within the green industries (*e.g., lighting systems, water features, irrigation systems, sawmill operation, sod farming, nursery production, dairy herd inspection*) and the qualifications required to work in these areas;
- D3.4** identify ways of acquiring knowledge and experience in green industry occupations (*e.g., through part-time work experience, cooperative education, guest speakers, field trips, job shadowing*);
- D3.5** identify the scope and strength of the labour force within the green industries regionally, nationally, and internationally;
- D3.6** demonstrate an understanding of and apply the Essential Skills that are important for success in the green industries, as identified in the Ontario Skills Passport (*e.g., reading text, writing, document use, computer use, oral communication, numeracy, thinking skills*);
- D3.7** demonstrate an understanding of and apply the work habits that are important for success in the green industries, as identified in the Ontario Skills Passport (*e.g., initiative, ethical conduct, accountability, entrepreneurship, customer service*);
- D3.8** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in areas related to the green industries (*e.g., Passport to Safety certificate, reports, assignments, Ontario Skills Passport Work Plan*), and explain why having a current portfolio is important for career development and advancement.